

Weiyao Tao

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EDUCATION

Johns Hopkins University <i>M.S.E. in Biomedical Engineering (THU-JHU Dual Degree Program)</i>	Baltimore, USA Aug. 2025 – June 2026
Tsinghua University <i>M.S. Candidate in Electrical Engineering (THU-JHU Dual Degree Program)</i>	Beijing, China Aug. 2026 – June 2028
University of Oxford <i>Visiting Student, Department of Computer Science, St Anne's College</i>	Oxford, England Oct. 2023 – Jul. 2024
Xi'an Jiaotong University <i>B.Eng. in Automation (Honors Program) and B.Eng. in Information Engineering</i>	Xi'an, China Sep. 2021 – Jul. 2025

TECHNICAL SKILLS

Programming Languages: Python (Pytorch, Tensorflow), C/C++, Matlab
Developer Tools: Git, Jupyter Notebook, Anaconda, Docker, VS Code, Visual Studio, PyCharm, Keil
English Proficiency: IELTS 8.0

EXPERIENCE

Researcher, Surgical Workflow Learning via Time-to-Event Modeling <i>Johns Hopkins University</i>	Jan. 2026 – Aug. 2026 Baltimore, USA
- Developed SurgTTE, an online framework for surgical workflow anticipation that jointly recognizes the current surgical step and predicts which step occurs next and when the transition will occur. - Reformulated surgical workflow anticipation from deterministic remaining-time regression as discrete time-to-event modeling, learning a competing-risks joint probability distribution over next-step identity and transition time, together with a time-to-end distribution when completion labels are available. - Evaluated SurgTTE across three cataract surgery datasets and Cholec80, showing consistent gains over regression-based baselines (improved next-step accuracy by up to 14.6 percentage points and reduced long-horizon timing error by up to 26.9%). - Derived single-pass confidence estimates directly from predicted event-time distributions and evaluated them through selective prediction, confidence-error analysis, and probabilistic calibration, without test-time sampling or ensembling. - Manuscript <i>SurgTTE: Distributional Time-to-Event Modeling for Online Surgical Workflow Anticipation</i> under review at WACV 2027 Applications Track.	
Researcher, Uncertainty Quantification for Survival Analysis <i>University of Oxford</i>	Jan. 2024 – Aug. 2024 Oxford, England
- Developed survival prediction models using Deep Learning (DeepHit, DeepSurv, Deep Survival Machine), Machine Learning (Random Survival Forest), and Bayesian methods (BNNSurv); implemented advanced uncertainty quantification techniques (MC-Dropout, Deep Ensembles, Bayesian methods). Proposed SurvUnc, a novel meta-model-based framework that quantifies prediction uncertainty for any survival model using an anchor-based learning strategy, showing significant improvements (up to 39.7%) in misprediction detection on four public datasets. Further proposed a comprehensive uncertainty quantification evaluation framework. - Authored and published <i>SurvUnc: A Meta-Model Based Uncertainty Quantification Framework for Survival Analysis</i>, accepted at KDD 2025. Open-sourced code at GitHub.	
Consulting Analyst, Global Consulting Project <i>Johns Hopkins University Global Consulting Project</i> <i>Client: International Iberian Nanotechnology Laboratory (INL)</i>	Jan. 2026 – Mar. 2026 Hybrid
- Advised the Portugal national lab client on commercialization strategy for a novel catalyst ink technology, integrating market, IP, cost, and scale-up analysis into client-facing recommendations. - Conducted patent and freedom-to-operate review across relevant filings, and developed a practical risk assessment and IP timeline to support positioning, disclosure planning, and next-step decisions. - Evaluated manufacturing readiness, raw-material economics, and partnership options, and recommended pilot and scale-up pathways to support qualification and market entry.	

Intern, Robot Products Department

June 2024 – July 2024

Nanjing Panda Electronics

Nanjing, China

- Engineered computer vision model (YOLO) for garbage classification, integrating real-time data acquisition and prediction.
- Developed PyQt UI for robot control, improving operational efficiency.

Undergraduate Researcher, Power System Load Forecasting

Mar. 2023 – Jun. 2023

Xi'an Jiaotong University

Xi'an, China

- Reviewed statistical and machine learning strategies for short-term and long-term power system load forecasting.
- Implemented LSTM and deep learning architectures to predict short-term power load.

ACTIVITIES

Executive Committee Member

Oct. 2023 – Jun. 2024

Oxford University Chinese Students and Scholars Association

Oxford, England

- Designed and executed major events, such as the Oxford Freshman Ball, Entrepreneurship Competition, and Dragon Spring Gala, enhancing student engagement.

Department Head & Team Leader & Volunteer

Oct. 2021 – Aug. 2022

Yingzai Assistance and Care Society, Xi'an Jiaotong University

Xi'an, China

- Led large-scale volunteer activities, including weekly teaching for rural students with 1000+ participants.
- Organized and led summer student practice teams in Lantian County, Shaanxi (2022) and Yushu Tibetan Autonomous Prefecture, Qinghai (2023); received multiple national awards for rural education support.

Volunteer

Jan. 2024 – Feb. 2024

ITCILO 2024 Global Youth Forum

Online

- Engaged in discussions on work, employment, and social issues globally;
- Collaborated with youth and stakeholders to share perspectives and contribute to policy formulation, empowering young people to address social challenges and shape their futures through workshops and networking opportunities.

AWARDS & HONORS

- **Outstanding Winner** (~0.18% winning rate), American Mathematical Contest in Modeling (2023)
- **National First Prize**, 14th National Mathematics Competition for College Students (2022)
- **International Third Prize** (rank 12/2867), IKCEST 4th "One Belt, One Road" International Big Data Competition (2022)
- **Scholarships and Honors:** Shuiyou First Class Scholarship (2021–2022), Qiu Changrong First Class Scholarship (2022–2023); Outstanding Student (2021, 2022, 2023), Outstanding Graduate (2025), *Xi'an Jiaotong University*